Marmara University
IE3081 – Modeling and Discrete Simulation Course
Fall 2021 Term Project

SYSTEM

I built a simulation for a clinic with one doctor. Patients may have appointment in advance, or they can register in clinic. Patients with age less than or equal to seven and greater or equal to sixty-five has more priority in examination queue than others. It has two queues, two delays (registration and examination). Examination time varies between five and thirteen minutes. Examination delay of patients in prioritized age group takes 25% longer than others. Patient age distribution calculated from Turkey's population pyramid. Patient priority inversely proportional with arrival time. Patient has appointment count is fixed, 25. Outpatient rate normally distributed and varies between 0.40 and 0.60.

Patient priority calculated with this formula:

$$Patient\ priority = \frac{1}{time()}$$

Patient priority for prioritized patients calculated with this formula:

$$Patient\ priority\ = \frac{1}{2 \cdot time()}$$

Outpatient (without appointment) count calculated with this formula:

$$Outpatient\ count = \frac{25 \cdot Outpatient\ rate}{1 - Outpatient\ rate}$$

SYSTEM COMPONENTS

Entities

- Patient
- Hospital receptionist
- Doctor

Attributes

- Patient's appointment method
- Processing speed of receptionist
- Processing speed of doctor

Activities

- Waiting in registration queue
- Register for examination
- Waiting in examination queue
- Get examined

Events

- Arrival of patient
- Register of patient
- Examination of patient

State Variables

- Number of patients with appointment in advance
- Number of patients without appointment
- Number of patients waits for registration
- Number of patients waits for examination
- Number of patients got examined

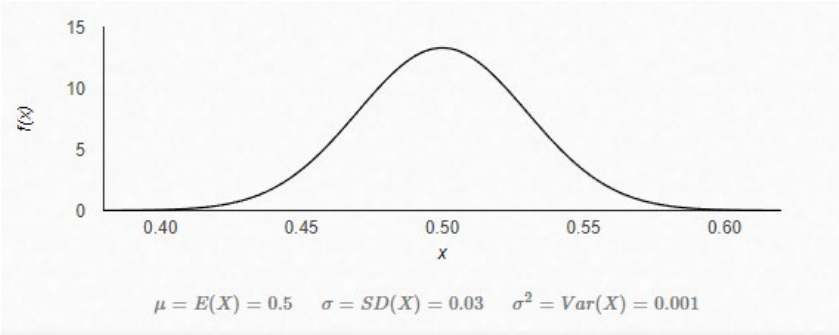
Relationship Between Components

When patient arrives if he has an appointment, he directly goes to examination queue. If he does not have appointment he needs to register first. So, he must go registration queue first and register for examination. After the registration he can go examination queue. Examination queue is not a FIFO queue, it is priority queue (patient priority). When he's turn come, he gets examined. After the examination he lefts the clinic.

INPUT PARAMETERS

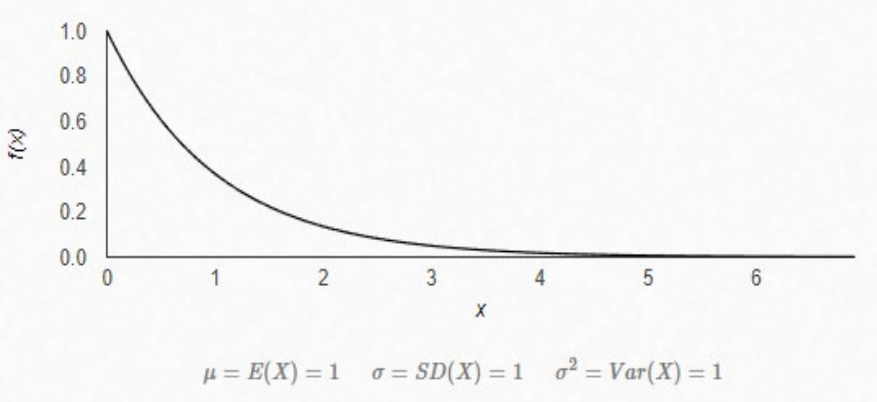
Patient without Appointment Rate

It is a random number from a normal distribution with mean 0.5 and sigma 0.03.



Interarrival Distribution of Patient without Appointment

It is an exponential distribution.



Interarrival Distribution of Patient with Appointment in Advance

Interval start	Interval end	Probability	Cumulative probability
10	10	0.20	0.20
11	12	0.20	0.40
13	13	0.20	0.60
14	14	0.20	0.80
15	15	0.20	1.00

Patient Age Distribution

Interval start	Interval end	Probability	Cumulative probability
0	4	0.0732	0.0732
5	9	0.0781	0.1513
10	64	0.7536	0.9049
65	10	0.0951	1.0000

Registration Delay Distribution

Interval start	Interval end	Probability	Cumulative probability
1	1	0.30	0.30
2	2	0.40	0.70
3	3	0.15	0.85
4	4	0.15	1.00

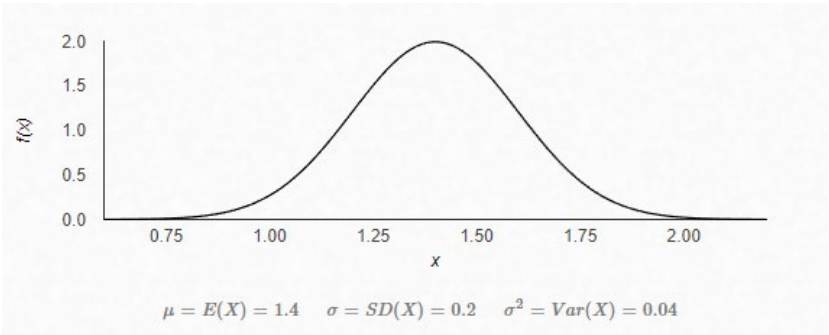
Examination Delay Distribution

Examination of patients in prioritized age group extended by 25%.

Interval start	Interval end	Probability	Cumulative probability
5	5	0.05	0.05
6	8	0.15	0.20
9	10	0.30	0.50
11	13	0.30	0.80
14	16	0.15	0.95
17	20	0.05	1.00

Patient Moving Speed Distribution

It is normally distributed, and unit is meters per second.



OUTPUT PARAMETERS

Sample size is 10 for all outputs.

BASE SYSTEM

Measurement	Value	Std. Dev.	CI
Average patient age	40.653 years	1.813144	1.12377744
Average number of patients has appointment	25	0	
Average number of patients without appointment	24.4	2.270585	1.40729721
Average wait time for registration	13.32 mins	5.464772	3.38703881
Average registration time	2.133 mins	0.174423	0.10810642
Average wait for examination	223.361 mins	24.9355	15.4548991
Average wait for examination (priority group)	221.6262 mins	49.14564	30.4602259
Average examination time	11.701 mins	0.570603	0.3536569
Length of stay	247.496 mins	26.73857	16.5724348
Wait time difference between priority group and others	-0.78%		

Narrowing CI 10%

Confidence interval of length of stay is 16,57 and we aim narrow it 10% (14,91).

Sample Size	CI	Difference from the target
11	15,8012	+0,8912
12	15,12849	+0,21849
13	14,53499	-0,37501

When sample size is 12 it gives the closest result to our aim.

Prediction Intervals

Measurement	Prediction Interval
Average patient age	40.653±4.30 years
Average number of patients has appointment	25±0
Average number of patients without appointment	24.4±5.39
Average wait time for registration	13.32±12.97 mins

Average registration time	2.133±0.41 mins
Average wait for examination	223.361±59.16 mins
Average wait for examination (priority group)	221.6262±116.60 mins
Average examination time	11.701±1.35 mins
Length of stay	247.496±63.44 mins

ALTERNATIVE SYSTEM

In the alternative system I separated queues for regular patients and prioritized patients. The doctor examines prioritized patients first if no one left continues with regular patients.

Measurement	**Value**	**Std. Dev.**	**CI**
Average patient age	40.187 years	3.287556	2.037611
Average number of patients has appointment	25	0	
Average number of patients without appointment	25.5	3.407508	2.111957
Average wait time for registration	13.73 mins	3.877685	2.40337
Average registration time	2.219 mins	0.15264	0.094605
Average wait for examination	221.795 mins	27.95338	17.32536
Average wait for examination (priority group)	9.6033 mins	4.321988	2.678747
Average examination time	11.492 mins	0.353075	0.218834
Length of stay	246.064 mins	28.91724	17.92276
Wait time difference between priority group and others	-95.67%		

Measurement	**Prediction Interval**
Average patient age	40.187±7.80 years
Average number of patients has appointment	25±0
Average number of patients without appointment	25.5±8.08
Average wait time for registration	13.73±9.20 mins
Average registration time	2.219±0.36 mins
Average wait for examination	221.795±66.32 mins
Average wait for examination (priority group)	9.6033±10.25 mins

Average examination time	11.492±0.84 mins
Length of stay	246.064±68.61 mins

Reduce Half Width of Confidence Interval

$$t_{\alpha/2} = 2.262157163$$

Measurement	**CI Half-Width**	**CI Half-Width Reduced**
Average patient age	2.351775767	2.11659819
Average number of patients has appointment	0	0
Average number of patients without appointment	2.437584416	2.193825974
Average wait time for registration	2.773928655	2.49653579
Average registration time	0.109191889	0.0982727
Average wait for examination	19.99664196	17.99697777
Average wait for examination (priority group)	3.091764273	2.782587845
Average examination time	0.25257491	0.227317419
Length of stay	20.68614695	18.61753225

Confidence interval half width of length of stay is 20.69 and we aim narrow it 10% (18.62).

Sample Size	**CI Half-Width**	**Difference from the target**
11	19.72347	+1.10347
12	18.88378	+0.26378
13	18.14295	-0.47705

When sample size is 12 it gives the closest result to our aim.

DIFFERENCE BETWEEN SYSTEMS

Measurement	CI Base	CI Alternative
Average patient age	1.12377744	2.037611
Average number of patients without appointment	1.40729721	2.111957
Average wait time for registration	3.38703881	2.40337
Average registration time	0.10810642	0.094605
Average wait for examination	15.4548991	17.32536
Average wait for examination (priority group)	30.4602259	2.678747
Average examination time	0.3536569	0.218834
Length of stay	16.5724348	17.92276

In generally confidence intervals of the alternative system is less than confidence intervals of the base system.

CI Average Base	CI Average Alternative
8.6084296	5.5991555

Confidence interval average of the alternative system is 34.95% less than confidence interval average of the base system.